Chengxiao Dai

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Google Scholar LinkedIn

Research Interests: Large Language Models & Multi-Agent Systems | Retrieval-Augmented Generation & Knowledge Graph Reasoning | LLM Safety & Copyright Protection

Education

The University of Sydney, Australia — Master of Computer Science (Advanced Entry) Aug 2025 – Present

Core Courses: Principles of Data Science, Machine Learning, Computational Statistical Methods, Data Engineering, Information Theory & Self-Organisation, Discrete Optimisation, Randomised & Advanced Algorithms.

Universiti Tunku Abdul Rahman, Malaysia — Bachelor of Computer Science Jan 2021 – Oct 2024

Core Courses: Object-Oriented Programming, Data Science, Algorithm Analysis, Operating Systems, Data Structures & Algorithms, Database Technology, Front-End Development, Cybersecurity.

Work (Internship) Experience

The University of Sydney — Tutor Feb 2026 – Present

- Tutor for ***DATA2001: Data Science, Big Data and Data Variety***, responsible for lab teaching of 3 classes totalling 65 students; primarily explaining core data science concepts, guiding students through the corresponding lab code, and assisting with exam invigilation.

Singapore Institute of Manufacturing Technology, Agency for Science, Technology and Research — Research Intern Oct 2025 – Jan 2026

- Contributed to building a **Text2SPARQL / GraphRAG** knowledge-graph question-answering system for circular-economy scenarios; modelled domain ontologies (industrial classification, waste types, resource matching, carbon-emission factors) on RDF / OWL, and designed **18 classes** of lightweight SPARQL query templates covering entity lookup, code alignment, multi-hop path discovery, and Top-K ranking. Integrated LLMs for schema alignment, entity linking, template selection, and slot filling, reducing field hallucination, relation-direction errors, and path-composition errors that arise from directly generating SPARQL | Project Technical Report.
- Designed and built a constrained reinforcement-learning multi-agent context-construction framework, abstracting subgraph expansion, path exploration, and evidence selection as a multi-agent sequential decision task. Designed the **LC-MAPPO** (Lagrangian-Constrained Multi-Agent PPO) algorithm to jointly optimise the policies of subgraph-architecture, path-navigation, and context-selection agents under the CTDE (centralised training, decentralised execution) paradigm; at inference, each agent independently executes discrete actions (continue, backtrack, adopt, stop) based on the current graph state. Under three constraints (edge expansion, interaction steps, and token cost), it achieved **+39.3 EM@1** (Top-1 exact-match) improvement over GraphRAG on multi-hop KGQA, while reducing end-to-end latency by **18.6%** and subgraph size by **40.9%**.

China Academy of Information and Communications Technology — AI Researcher Feb 2025 – Jun 2025

- Co-designed an LLM identity-tag embedding and attribution-verification scheme for copyright attribution and provenance tracing under model reuse, transfer, and re-fine-tuning scenarios. The scheme built a transferable identity-tag adapter on **LoRA**, designed trigger samples across **4 modalities** (text, image, speech, structured data), and bound input triggers to preset outputs to form an identity-tag dataset. Combining **module-importance evaluation**, multi-path redundant embedding, teacher-student distillation preservation, and a **Sigmoid gating activation** mechanism, it achieved transfer of identity tags between base and downstream LLMs; quantified by **Fingerprint Success Rate (FSR)**, it reached up to **100% FSR** under both Direct LoRA injection and Transfer settings. The scheme resulted in a **national invention patent** "Method, Apparatus, and Electronic Device for Embedding and Transferring Multimodal Controllable LLM Identity Tags Based on Low-Rank Fusion" (Publication No.: CN 121278693 A).
- Led frontier-technology research on **LLMs, multi-agent development frameworks, and model safety**, producing **30+** technical research reports, standard interpretations, and briefing materials. Participated in formulating **2 industry standards**, including the ***Maturity Assessment Standard for Power-Sector AI Semantic LLM Services***, responsible for standard-framework organisation, indicator design, and review-material preparation. Liaised with the **Baidu Qianfan LLM platform** to provide technical support and consulting for the deployment of **CNOOC Gas & Power Group's energy LLM**; supported **10+** relevant academic and industry conferences, preparing technical materials, training courseware, and conference briefings.

Fujian Imperial Vision Technology Group Co., Ltd. — Algorithm Engineer Dec 2024 – Feb 2025

- Contributed to domain fine-tuning of image-generation models based on **Stable Diffusion 1.5 / SDXL**, responsible for building portrait-style datasets: curated **4000+** portrait and style images, auto-labelled with BLIP-2 / WD14-Tagger and human-reviewed to generate captions; performed data cleaning via CLIP semantic similarity, Aesthetic Predictor, and pHash deduplication, removing **600+** low-quality, duplicate, and style-inconsistent samples; using **Diffusers + LoRA** under bf16 mixed precision and a cosine LR scheduler, ran **5+** hyperparameter comparison experiments, improving character identity consistency and generation stability for specific styles.
- Built an AIGC service inference gateway based on **FastAPI + Uvicorn multi-worker**, encapsulating **3+** core interfaces

for image generation, super-resolution restoration, and portrait restoration; introduced container-startup model pre-loading, a **Celery / Redis** asynchronous task queue, an aiohttp connection pool, and S3 Multipart concurrent chunked upload to optimise image-asset upload and async task scheduling, reducing interface cold-start time by **30%** and supporting stable scheduling of **20+ concurrent tasks**.

Ruijie Networks Co., Ltd. — Cloud Computing Operations Engineer Oct 2023 – Jan 2024

- Participated in daily operations and resource delivery of a private cloud platform, responsible for base-environment configuration of **20+** Linux servers, virtualisation resource inspection, cloud-host creation and resource allocation, and storage/network connectivity verification; processed **50+** resource-delivery and operations support tickets covering cloud-host creation, resource scaling, account-permission configuration, and service-fault troubleshooting, with average response time kept within **1 hour**.
- Wrote Shell / Python automation scripts for server-status checks, disk-space monitoring, service-process detection, and log archiving, covering routine inspection, alert troubleshooting, and document archiving; proactively identified **12** potential faults (low disk space, service anomalies, excessive resource usage) through automated inspection, assisting with resolution and documentation to ensure stable operation of the cloud platform's compute, storage, and network resources.

Research Experience

1. *From Knowledge to Inference: Scaling Laws of Specialized Reasoning on GlobalHealthAtlas* (Conference paper, **accepted at ICML 2026 – CORE A*/CCF-A**) | Paper Link.
2. *CLAUSE: Agentic Neuro-Symbolic Knowledge Graph Reasoning via Dynamic Learnable Context Engineering* (Conference paper, **accepted at ICLR – CORE A*/CCF-A**) | Paper Link.
3. *From Metaphor to Mechanism: How LLMs Decode Traditional Chinese Medicine Symbolic Language for Modern Clinical Relevance* (Conference paper, **accepted at IJCNN 2025 – CORE B/CCF-C**) | Paper Link.
4. *Multimodal Large Language Model Enhancement Network for Multimodal Sentiment Analysis* (Journal paper, **accepted at Multimedia Systems – JCR Q1 / IF 3.5**) | Paper Link.
5. *MEGA-RAG: A Retrieval-Augmented Generation Framework with Multi-Evidence Guided Answer Refinement for Mitigating Hallucinations of Large Language Models in Public Health* (Journal paper, **accepted at Frontiers in Public Health – JCR Q1 / IF 3.4**) | Paper Link.
6. *Multi-Stage Simulation of Residents' Disaster Risk Perception and Decision-Making Behavior: An Exploratory Study on Large Language Model-Driven Social–Cognitive Agent Framework* (Journal paper, **accepted at Systems – JCR Q1 / IF 3.1**) | Paper Link.

(2 IEEE SMC conference papers (CORE B/CCF-C), have been accepted. Additionally, 3 conference/journal papers are currently under review or in minor revision; currently **118** total citations and **H-index 5**.)

Competitions

- Kaggle | BirdCLEF+ 2026 | **Silver Medal (Rank: 143/4084, Top 4%)**
- Kaggle | LMSYS – Chatbot Arena Human Preference Predictions | **Silver Medal (Rank: 78/1849, Top 5%)**
- Kaggle | NeurIPS 2024 – Predict New Medicines with BELKA | **Bronze Medal (Rank: 107/1950, Top 6%)**
- Kaggle | Google – Fast or Slow? Predict AI Model Runtime | **Bronze Medal (Rank: 86/616, Top 14%)**

Honours & Awards

- **Youth Academic Paper Award**, Ministry of Industry and Information Technology, China – Oct 2025
- **Dean's List** – Sep 2024 Semester
- **Academic Improvement List** – Oct 2021 Semester & Jun 2022 Semester

Academic Service & Open-Source Community Contributions

- **Academic Service:** Reviewer and technical committee member for international conferences including **AAAI**, **ICIC**, **IJCNN**, **NeurIPS Workshop**, and **ICC**.
- **Open-Source Contributions:** Core contributor to the open-source projects **aiXiv** and **SkillBible**, totalling **100+** GitHub Stars.

Skills

- **Languages:** Fully English-taught at both bachelor's and master's levels; proficient in **English** for classroom teaching, academic reading, and paper writing; conversational **Malay** for daily communication.
- **Tech Stack:** LangChain / LangGraph, LlamaIndex, Hugging Face Transformers, PEFT (LoRA / QLoRA), MCP, FAISS, BM25, Sentence-Transformers, Cross-Encoder, RDF / OWL, SPARQL, GraphRAG, PyTorch, TensorFlow, Scikit-learn, Diffusers, Stable Diffusion / SDXL, TensorRT, FastAPI, Celery, Redis.
- **Programming Languages:** Python (Pandas, NumPy, PyTorch), R (ggplot2, dplyr), Java, C/C++, SQL, JavaScript, CSS.
- **Tools:** Linux, Docker, Git, AWS S3, VS Code, PyCharm, Jupyter Notebook, LaTeX (Overleaf), Microsoft Office.